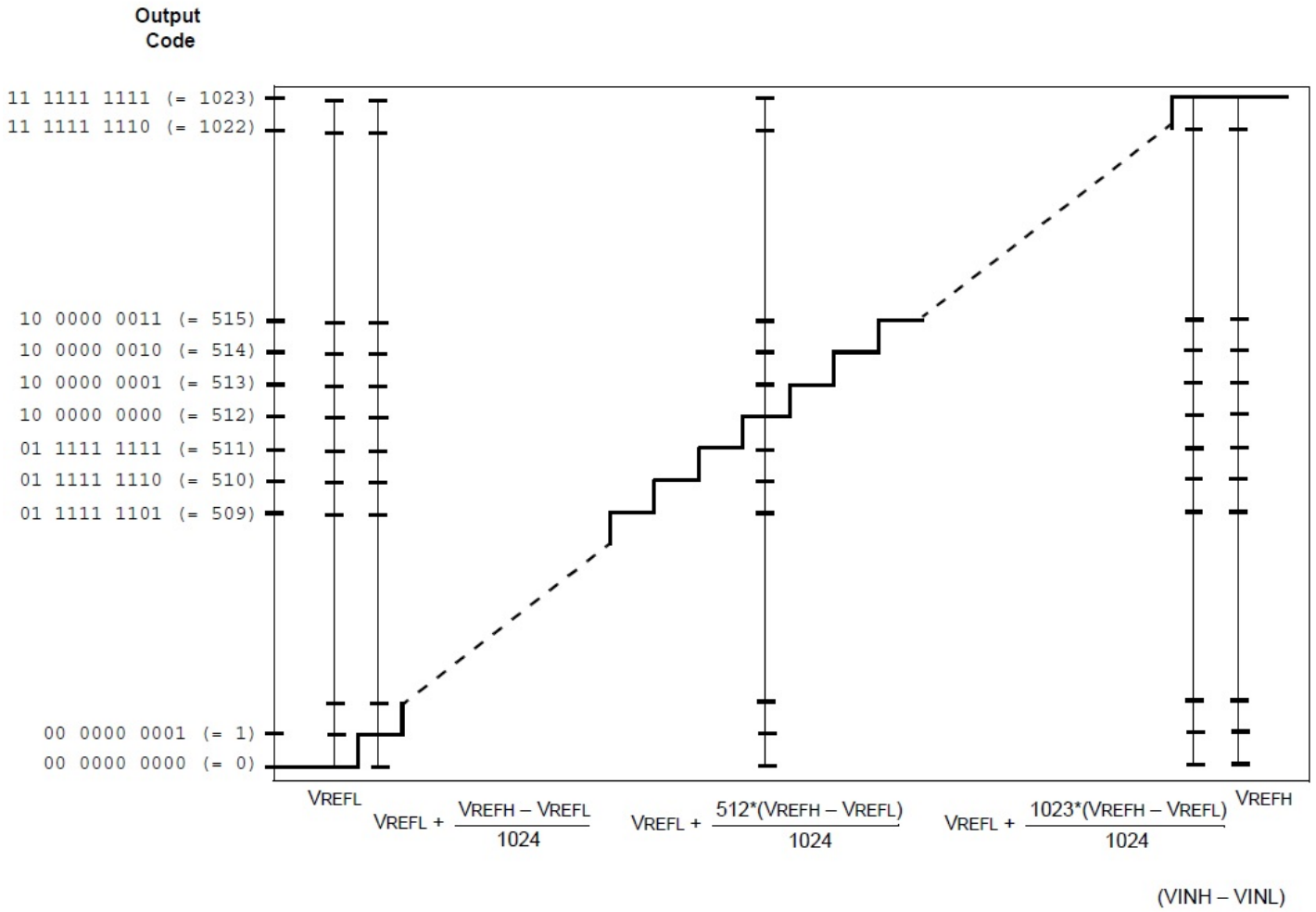
Università di Genova

Embedded Systems

ADC

Prof. Enrico Simetti

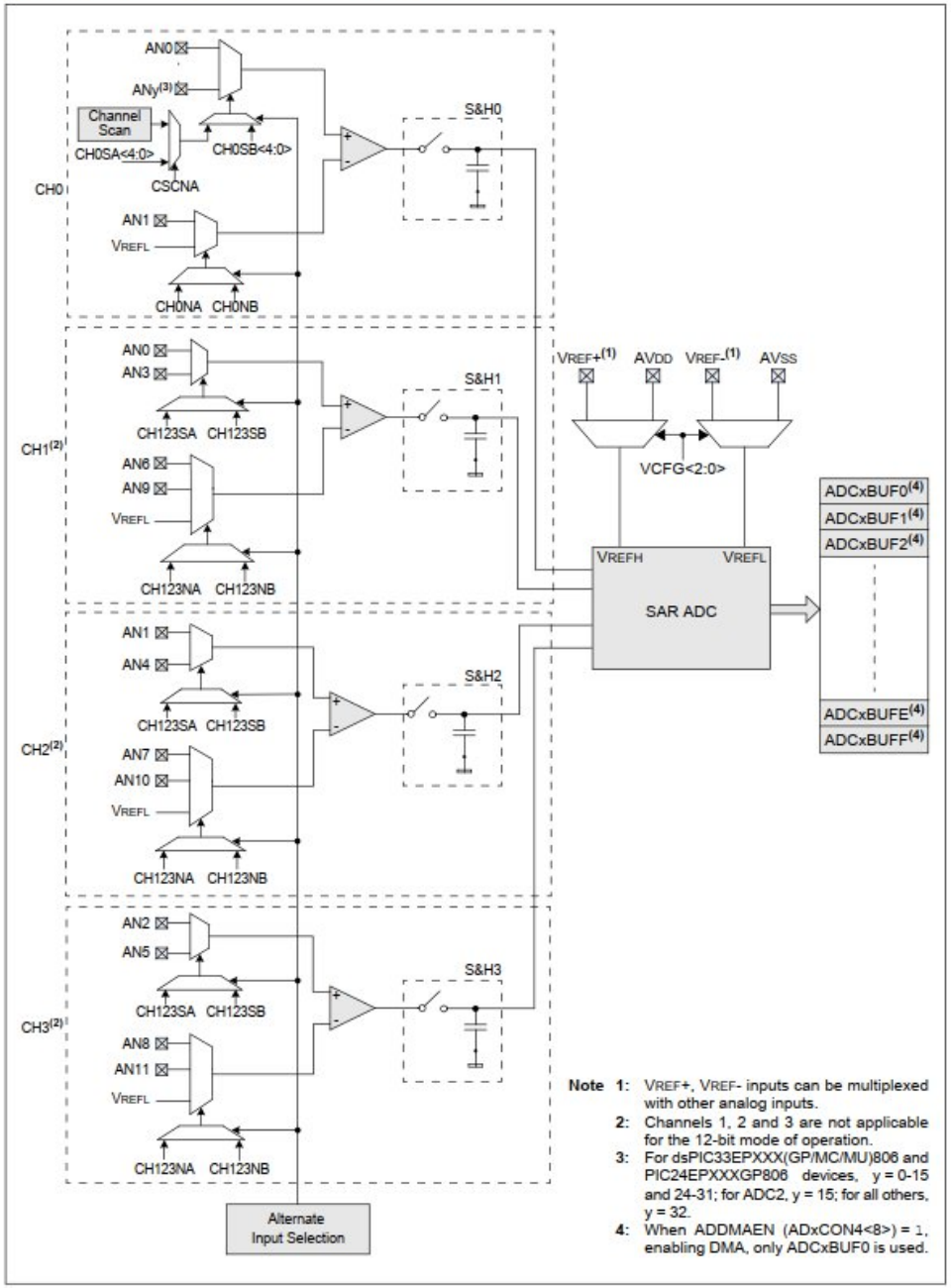
What Sampling Means



dsPIC33 ADC Modules

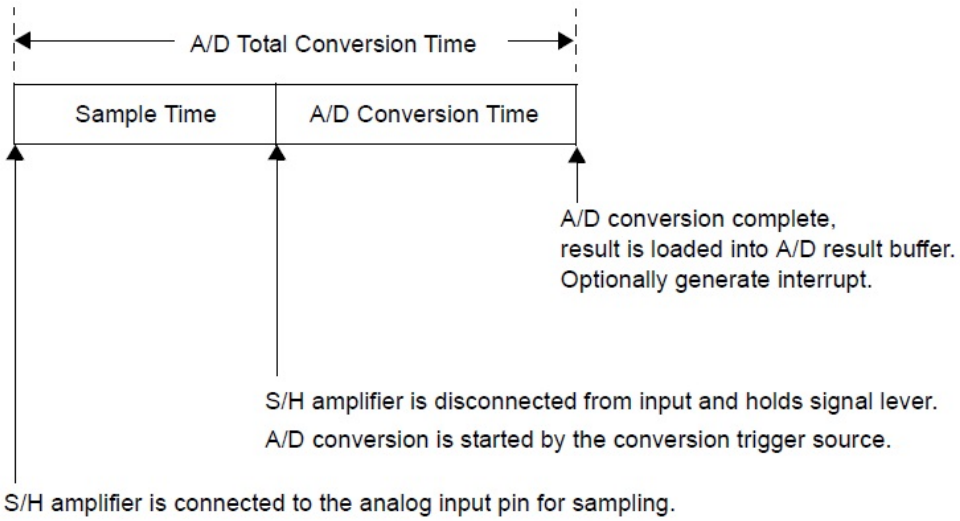
- Two ADC modules
- Up to 1 Msps
- 10-bit or 12-bit operation modes
- Up to 32 analog pins (ADC 1)
- Up to 16 analog pins (ADC 2)
- 4 differential S/H amplifiers (channels)
- 1 converter
- Channel scan mode
- 16 word buffer for conversion results
- Selectable conversion trigger source

FIGURE 23-1: ADCx MODULE BLOCK DIAGRAM



Note 1: VREF+, VREF- inputs can be multiplexed with other analog inputs.
2: Channels 1, 2 and 3 are not applicable for the 12-bit mode of operation.
3: For dsPIC33EPXXX(GP/MC/MU)806 and PIC24EPXXXGP806 devices, y = 0-15 and 24-31; for ADC2, y = 15; for all others, y = 32.
4: When ADDMAEN (ADxCON4<8>) = 1, enabling DMA, only ADCxBUF0 is used.

Sampling and Conversion



- Tad is ADC time period
- Tad duration is user defined [1 Tcy – 64 Tcy]
- Conversion time duration (Tconv): fixed to 12 Tad
- Sample time duration (Tsamp) can be
  - fixed (specified in number of Tads [0 Tad - 31 Tad])
  - determined by user software (by clearing SAMP bit)
- Sampling rate = 1/(Tconv+Tsamp)

For correct A/D conversions, the A/D conversion clock (Tad) must be selected to ensure a minimum Tad time of 154 nsec (for VDD = 5V)

ADC Triggers

Sampling period start

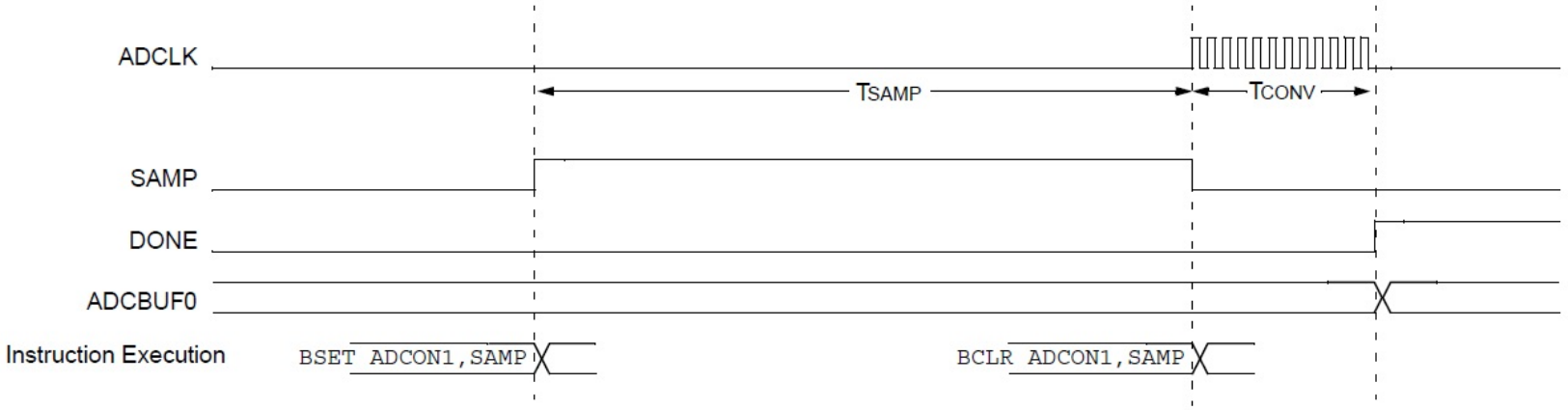
- automatic (at the end of previous conversion)
- manual (by setting SAMP bit in software)

Sampling period end (=conversion time start)

- automatic (if Tsamp is fixed)
- manual (by setting SAMP bit in software)
- (driven by other events)

ADC Triggers

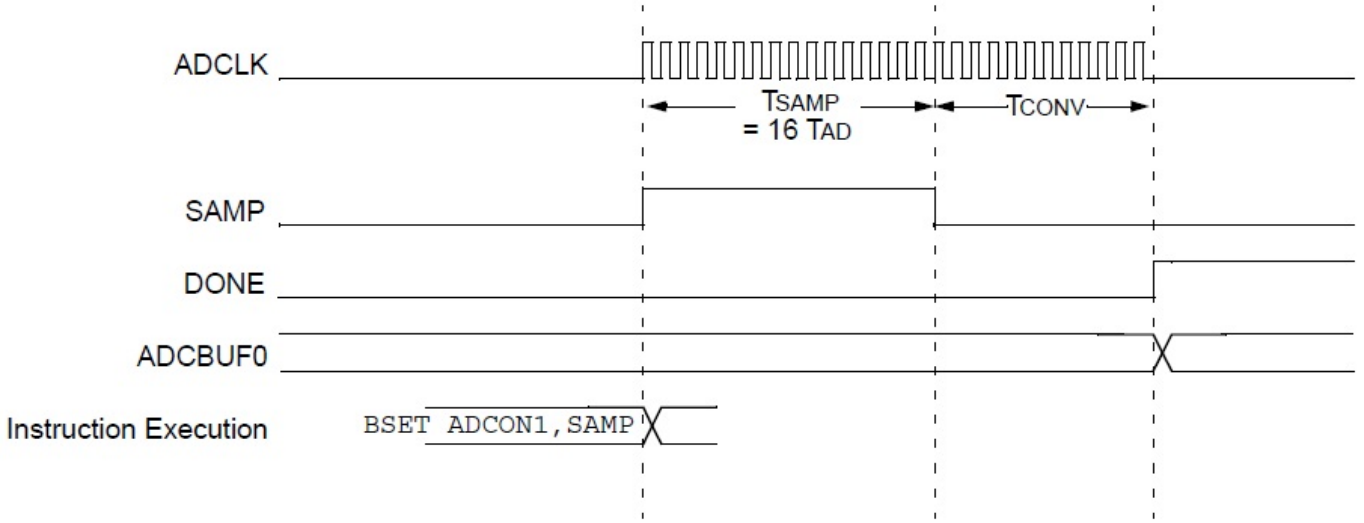
manual start - manual end




```
ADxCON1bits.ASAM = 0;  // x = 1 or 2, specifies ADC module
ADxCON1bits.SSRC = 0;
```

ADC Triggers

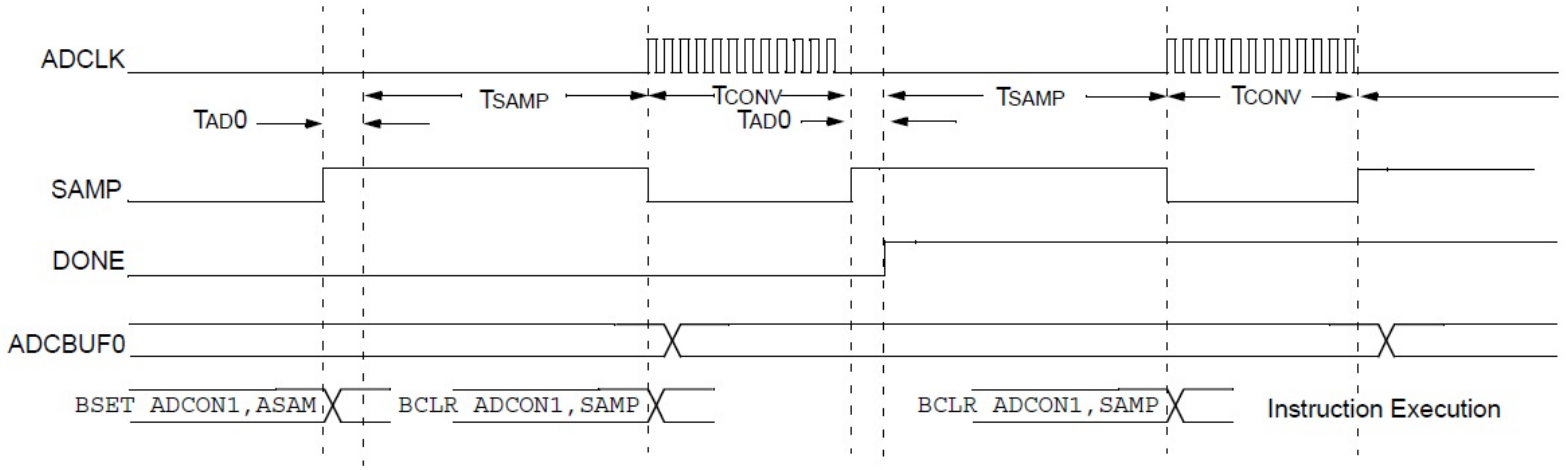
manual start - automatic end




```
ADxCON1bits.ASAM = 0;
ADxCON3bits.SAMC = 16; // sample time 16 Tad
ADxCON1bits.SSRC = 7; // conversion starts after time specified by SAMC
```

ADC Triggers

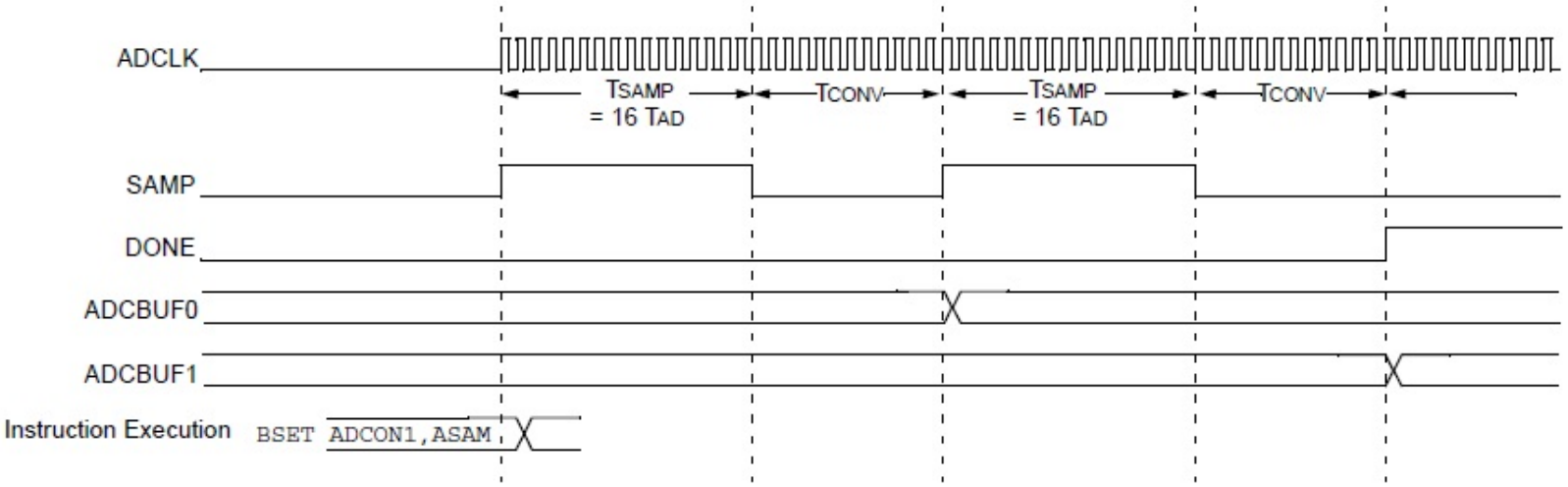
automatic start - manual end




```
ADxCON1bits.ASAM = 1;
ADxCON1bits.SSRC = 0;
```

ADC Triggers

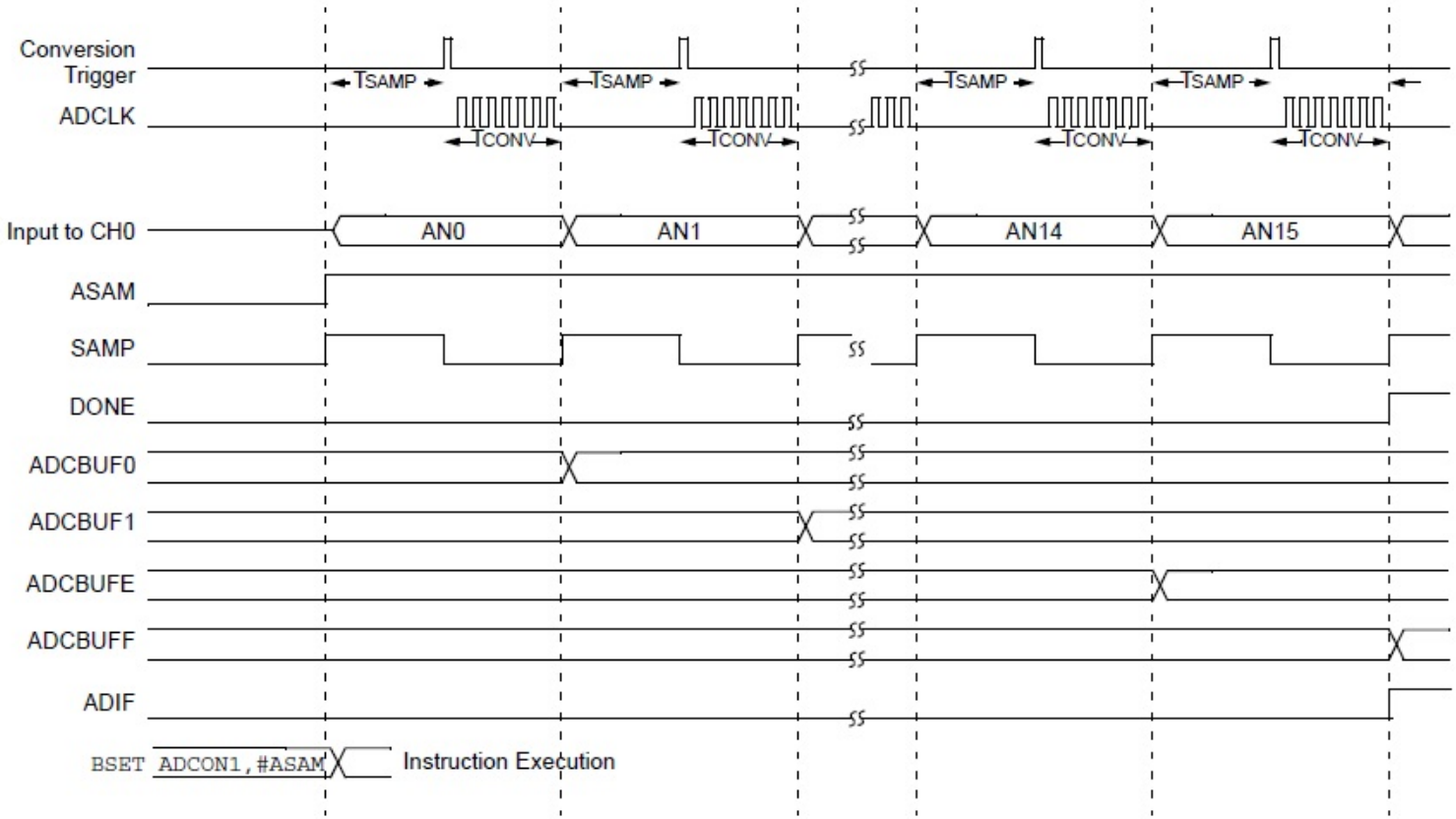
automatic start - automatic end




```
ADxCON1bits.ASAM = 1;
ADxCON3bits.SAMC = 16; // sample time 16 Tad
ADxCON1bits.SSRC = 7; // conversion starts after time specified by SAMC
```

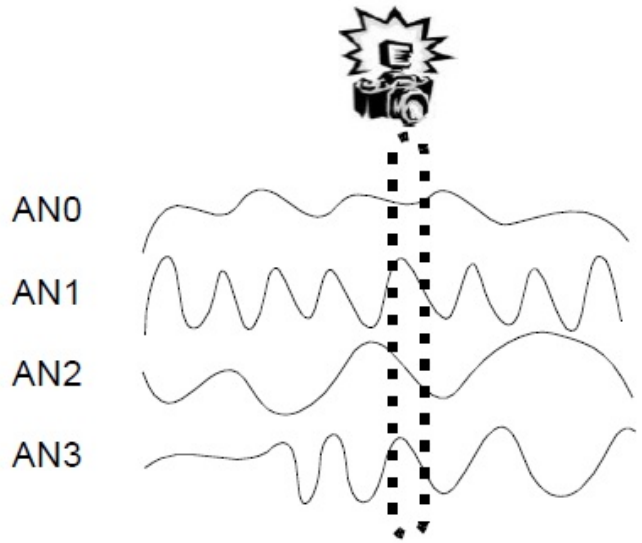
ADC Triggers

Scan mode

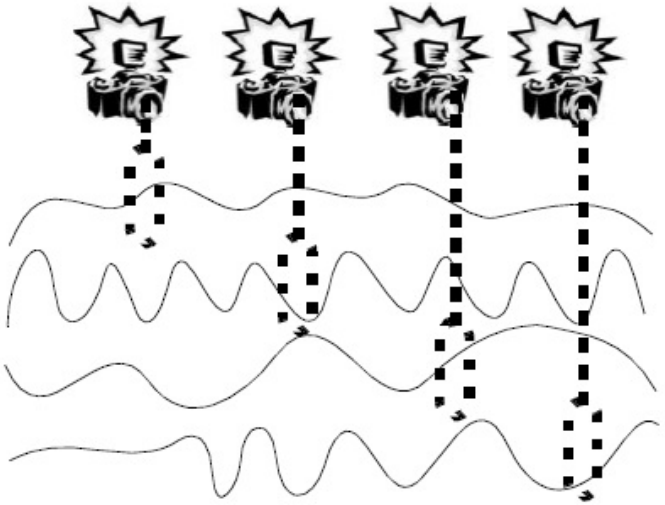



```
ADxCON2bits.CSCNA = 1; // scan mode
```

ADC Multiple Channels



Simultaneous Sampling



Sequential Sampling

ADxCON1 Register

REGISTER 23-1: ADxCON1: ADCx CONTROL REGISTER 1

R/W-0	U-0	R/W-0	R/W-0	U-0	R/W-0	R/W-0	R/W-0
ADON	—	ADSIDL	ADDMABM	—	AD12B(1)	FORM<1:0>	
bit 15							bit 8

R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0, HSC	R/C-0, HSC
SSRC<2:0>			SSRCG	SIMSAM	ASAM(3)	SAMP	DONE(3)
bit 7							bit 0

ADxCON1: ADC Control register 1.

- ADON: turns on the ADC module
- SSRC: selects how the conversion should start (0 = manual, 7 = internal counter)
- ASAM: selects how the sampling should start
- SAMP: sample enable bit
- DONE: tells when the A/D conversion is done

ADxCON2 Register

REGISTER 23-2: AD1CON2: ADC1 CONTROL REGISTER 2

R/W-0	R/W-0	R/W-0	U-0	U-0	R/W-0	R/W-0	R/W-0
VCFG<2:0>			—	—	CSCNA	CHPS<1:0>	
bit 15							bit 8

R-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0
BUFS	SMPI<4:0>					BUFM	ALTS
bit 7							bit 0

ADxCON2: ADC Control register 2:

- **CSCNA**: selects if inputs should be scanned
- **CHPS**: selects the channels
- **SMPI**: after how many conversions when the IF should be set to 1

ADxCON3 Register

REGISTER 23-4: ADxCON3: ADCx CONTROL REGISTER 3

R/W-0	U-0	U-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0
ADRC	—	—	SAMC<4:0>(1)				
bit 15							bit 8

R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0
ADCS<7:0>(2,3)							
bit 7							bit 0

ADxCON3: ADC Control register 3:

- SAMC: how long the sampling should last [0 Tad - 31 Tad] (it matters only if SSRC = 7)
- ADCS: selects how long is one Tad [1 Tcy - 64 Tcy]

ADxCHS0 Register

REGISTER 23-7: ADxCHS0: ADCx INPUT CHANNEL 0 SELECT REGISTER

R/W-0	U-0	U-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0
CH0NB	—	—	CH0SB<4:0>(1)				
bit 15							bit 8

R/W-0	U-0	U-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0
CH0NA	—	—	CH0SA<4:0>(1)				
bit 7							bit 0

ADxCHS0: selects the inputs to channel 0

ADxCHS123 Register

REGISTER 23-6: ADxCHS123: ADCx INPUT CHANNEL 1, 2, 3 SELECT REGISTER

U-0	U-0	U-0	U-0	U-0	R/W-0	R/W-0	R/W-0
—	—	—	—	—	CH123NB<1:0>		CH123SB
bit 15							bit 8

U-0	U-0	U-0	U-0	U-0	R/W-0	R/W-0	R/W-0
—	—	—	—	—	CH123NA<1:0>		CH123SA
bit 7							bit 0

ADxCHS123: selects the inputs to channels 1, 2 and 3

Results will be found in registers ADCxBUF0 through ADCxBUFF

Other Registers

Register 16-7: AD1CSSH: ADC1 Input Scan Select Register High

R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0
CSS31	CSS30	CSS29	CSS28	CSS27	CSS26	CSS25	CSS24
bit 15							bit 8

R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0
CSS23	CSS22	CSS21	CSS20	CSS19	CSS18	CSS17	CSS16
bit 7							bit 0

Register 16-8: ADxCSSL: ADCx Input Scan Select Register Low

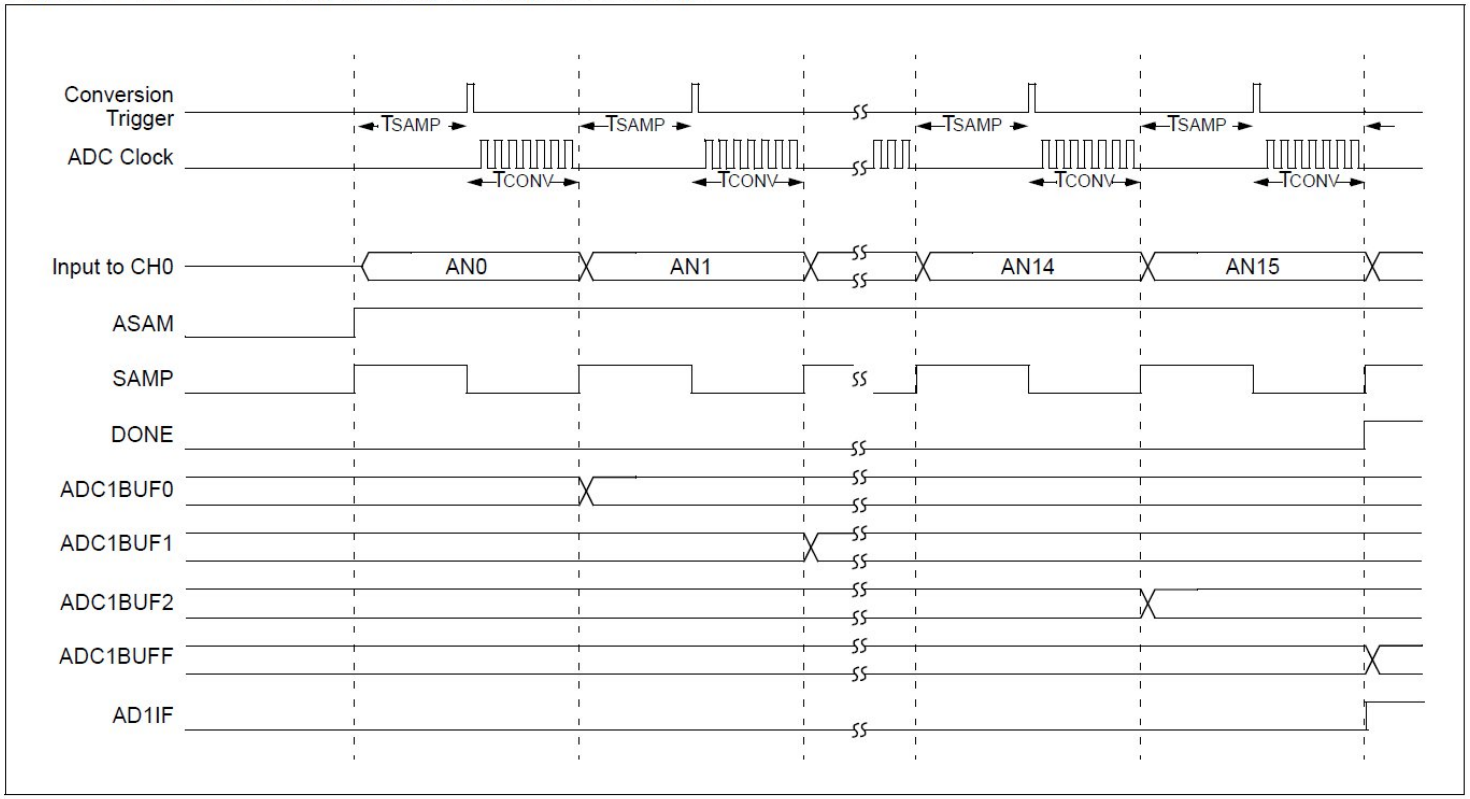
R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0
CSS15(3)	CSS14(3)	CSS13(3)	CSS12	CSS11	CSS10	CSS9	CSS8
bit 15							bit 8

R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0
CSS7	CSS6	CSS5	CSS4	CSS3	CSS2	CSS1	CSS0
bit 7							bit 0

1 = scan, 0 = do not scan

Scan Mode

Figure 16-27: Scanning Through 16 Inputs/Interrupt



ANSELx Register

ANSELx: sets pins as analog (1) or digital (0). As a thumb rule, always set each ANSELx to 0 and then toggle to 1 only the pins required to be analog

These registers are important, failing to configure them will result in errors during the conversion

Configuring the ADC

1. select your Tad ADC time and set ADxCON3bits.ADCS (suggested value is 8 for our board)
2. select how sampling begins
   - manual: ADxCON1bits.ASAM = 0;
   - automatic: ADxCON1bits.ASAM = 1;
3. select how sampling ends and conversion begins
   - manual: ADxCON1bits.SSRC = 0;
   - automatic: ADxCON1bits.SSRC = 7; and then choose how long sampling lasts by setting ADxCON3bits.SAMC;
4. choose how many channels you want to use setting ADxCON2bits.CHPS
5. choose the positive input to the channels by setting ADxCHS0bits.CH0SA; and ADxCHS123bits.CH123SA;
6. set the input pins as analog (value 1) in ANSELx and the rest as digital (value 0)
7. if more than 1 channel is used, choose sequential or simultaneous sampling using AxDCON1bits.SIMSAM
8. if sequential set ADxCON2bits.SMPI; to number of inputs - 1; if simultaneous consider the number of channels used
9. finally, turn ADC on with ADxCON1bits.ADON = 1;

DONE bit

The DONE bit is only cleared at the start of each conversion and is set at the completion of the conversion, but remains set indefinitely, even through the next sample phase until the next conversion begins. If application code is monitoring the DONE bit in any kind of software loop, the user must consider this behavior because the CPU code execution is faster than the ADC.

As a result, in Manual Sample mode, particularly where the user's code is setting the SAMP bit, the DONE bit should also be cleared by the user application just before setting the SAMP bit.

IR Distance Sensor Specs

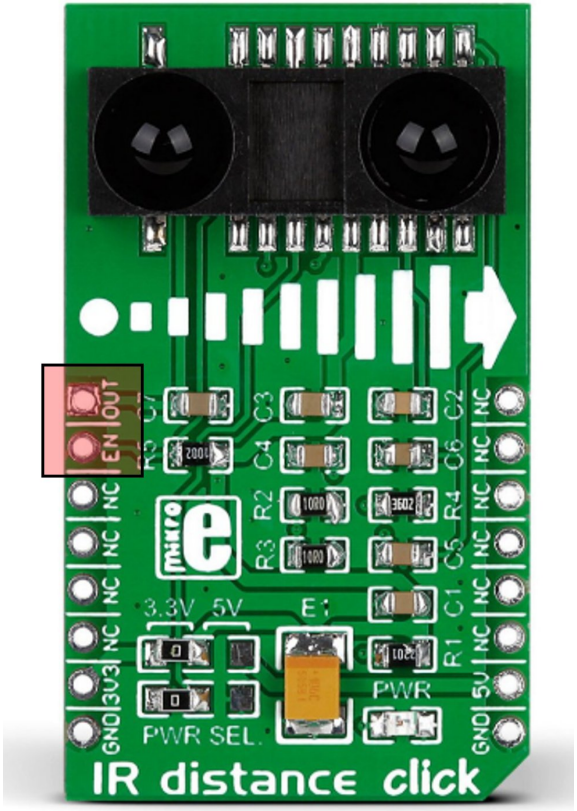
Infra-red distance sensor has a valid range of 10 cm to 1.5 m.

Output voltage on analog pin is inversely proportional to distance.

The EN pin is required to be high for the device to work properly

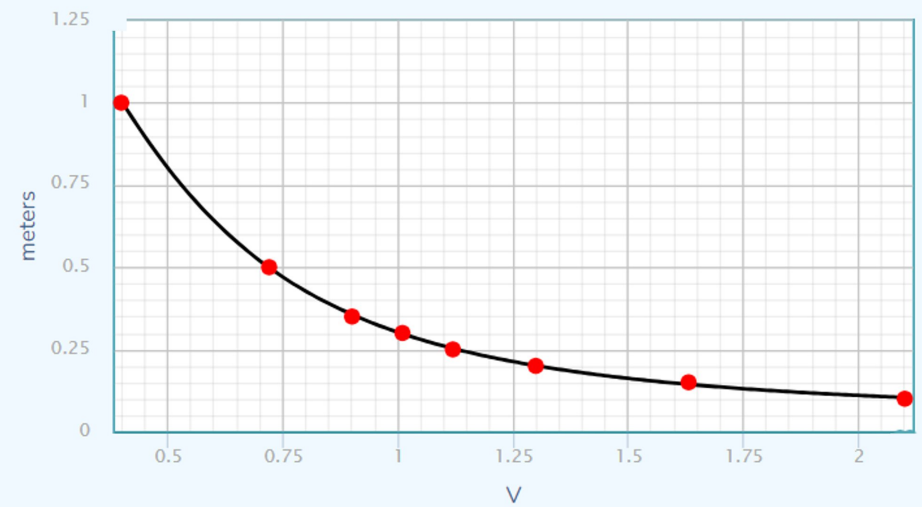
Measured voltage at 10 cm is (approximately) 2.1 V, at 1 m is 0.4 V.

Sensor response is nonlinear.



IR Distance Sensor Response

It can be approximated well with a fourth order polynomial

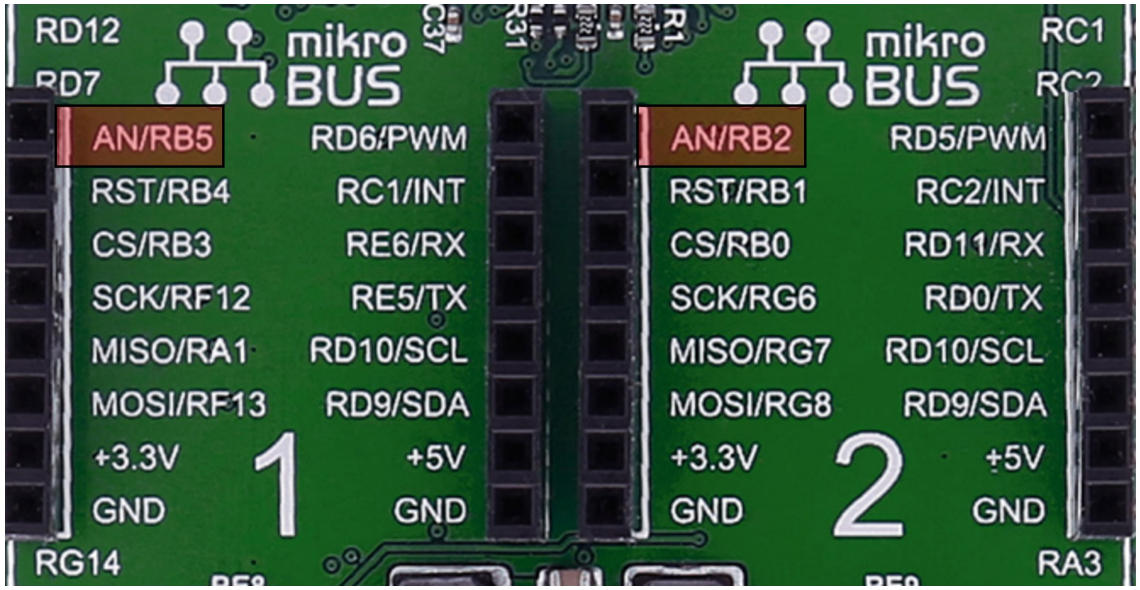


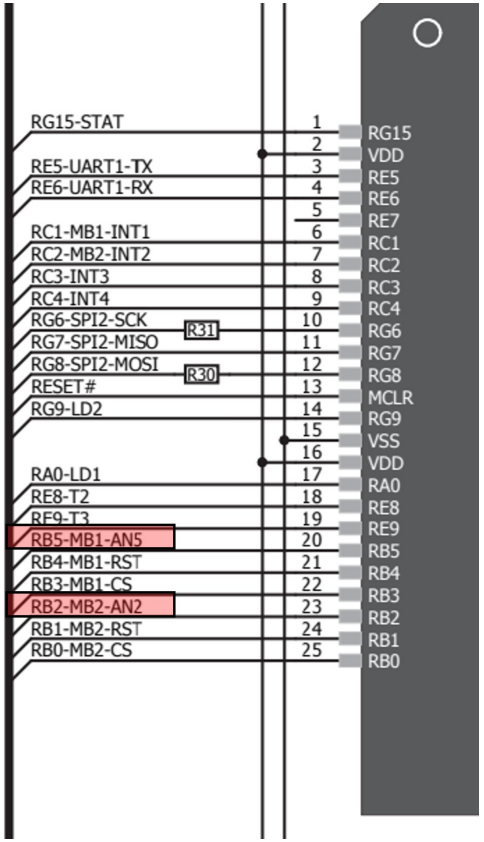
Given a voltage reading V, the distance measured by the sensor can be computed as:

$y = 2.34 - 4.74V + 4.06V^2 - 1.60V^3 + 0.24V^4$

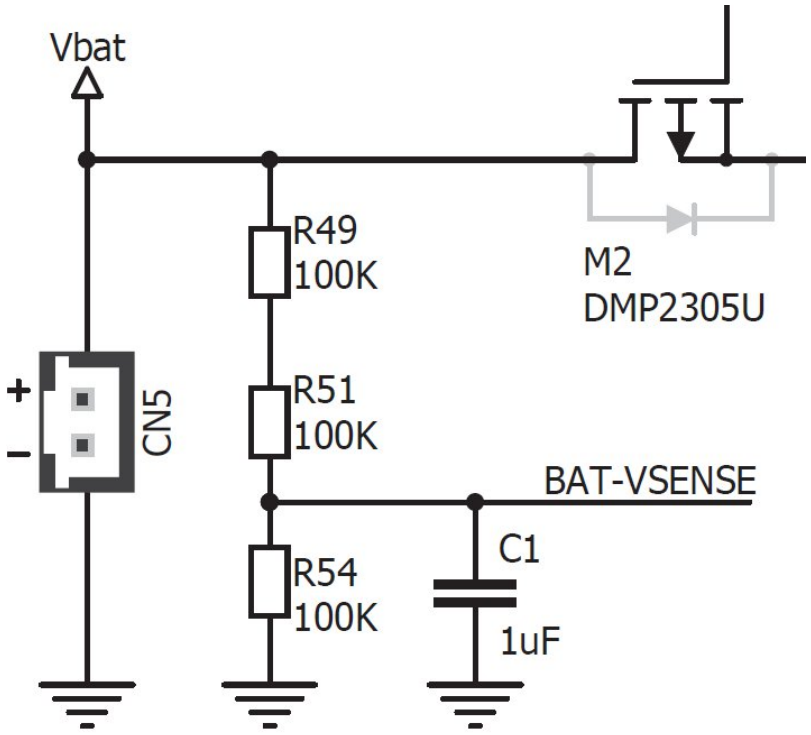
Analog Pins on dsPIC33

The MIKROBus sockets on the Clicker 2 board are connected to RB5 and RB2, corresponding to AN5 and AN2 on the dspic33 layout.





Battery Sensing



The BAT-VSENSE line is connected to AN11.

Assignment

- Read the value of the battery, considering the voltage divider circuit, and send it on the UART. Use manual sampling and manual conversion.
- Plug the IR Distance Sensor and the RS232 UART shield in the MIKRObus sockets of the Clicker board. Read the value of the distance sensor and send it on the UART. Use manual sampling and automatic conversion.
- Plug the IR Distance Sensor and the RS232 UART shield in the MIKRObus sockets of the Clicker board. Read both values using scan mode automatic sampling and conversion at 1 kHz. Send the values to the UART at 10 Hz using \$SENS,x,y* where x is the IR in cm, and y is the batter voltage in Volt.



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